Wheat Seed Variety Classification Using Deep Learning: A Comparative Study of Pre-trained, Custom, and Attention-Enhanced CNN Architectures

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***Abstract* —** Accurate classification of wheat seed varieties is critical for agricultural quality control, yield estimation, and automated grain sorting. This paper presents a structured deep learning investigation on a high-resolution RGB wheat seed dataset comprising 375 images across three varieties: Akbar-19, Dilkash-20, and Urooj-22. We systematically evaluate and compare five model configurations: three fine-tuned pre-trained CNNs (EfficientNetB0, ResNet50, VGG16), one custom CNN trained from scratch, and one attention-enhanced CNN incorporating a Squeeze-and-Excitation (SE) block. All models are evaluated under identical training conditions using accuracy, precision, recall, F1-score, AUC, confusion matrix, and statistical paired t-tests. Results show that VGG16 achieves the highest accuracy (55.65%) among the pre-trained models, while the custom CNN significantly improves performance (77.67% accuracy, AUC 0.9019). The attention-enhanced CNN achieves the best overall performance with 78.12% accuracy and the highest AUC of 0.9197, along with improved computational efficiency. This study highlights the challenges of training deep CNNs on small agricultural datasets and demonstrates that lightweight architectures and attention mechanisms can effectively improve performance under limited data conditions.

Index Terms *— Wheat seed classification, convolutional neural network, transfer learning, squeeze-and-excitation block, data augmentation, image classification.*

I. INTRODUCTION

Wheat is one of the world's most important staple crops, and accurate varietal identification is essential for maintaining seed purity, optimizing agricultural practices, and ensuring crop quality. Traditional wheat seed classification relies on manual inspection by trained agronomists, a process that is inherently time-consuming, subjective, and prone to human error. With the advent of high-resolution digital imaging and deep learning, automated image-based classification has emerged as a powerful alternative.

Convolutional Neural Networks (CNNs) have demonstrated exceptional performance in image classification tasks across numerous domains. In agricultural applications, CNNs and their variants have been applied to plant disease detection, fruit grading, and crop variety identification. However, agricultural datasets are often small in scale, posing significant challenges for data-hungry deep learning models.

This paper investigates the effectiveness of multiple deep learning strategies for classifying three wheat seed varieties — Akbar-19, Dilkash-20, and Urooj-22 — from a high-resolution RGB image dataset containing only 375 images. We evaluate: (1) transfer learning using three pre-trained CNNs from the literature, (2) a custom CNN designed and trained from scratch, and (3) an attention-enhanced CNN incorporating a Squeeze-and-Excitation (SE) block for adaptive feature recalibration. Our contributions are as follows:

- A structured comparative evaluation of five CNN architectures on a small-scale wheat seed dataset.
- Application of six augmentation strategies tailored specifically for wheat seed imagery.
- Integration of a Squeeze-and-Excitation attention mechanism into a custom CNN backbone.
- Statistical validation of all performance differences using paired t-tests.
- Identification of key challenges and practical recommendations for deep learning on small agricultural datasets.

II. RELATED WORK

Automated wheat seed classification using image processing and machine learning has been an active area of research. Table I summarizes key prior works on wheat seed and grain classification from reputed publishers.

TABLE I. Related Work Summary

Ref	Title (Short)	Dataset	Method	Accuracy	Key Finding
[1]	Wheat variety DL classification	Custom NIR (24 varieties, Turkey)	CNN on reflectance images	99.52%	High acc. on NIR-generated images
[2]	ML classification of wheat grains	Custom RGB (fresh vs rotten)	SVM, KNN, MLP, NB	~93%	SVM best on color features
[3]	Hybrid CNN hyperspectral	Custom (8 varieties, ~8000 images)	1D+2D CNN hybrid	95.65%	Spectral+spatial fusion effective
[4]	CNN & transfer learning wheat	Custom (124 varieties)	DenseNet201, MobileNet, InceptionV3	95.40%	DenseNet201 outperforms others
[5]	Wheat grain CNN classification	Custom (15 varieties, 15000 images)	CNN from scratch	97.53%	A large dataset enables high acc.
[6]	CNN-SVM bread wheat seeds	Custom wheat grain images	CNN + SVM classifier	Reported	Deep features + SVM robust
[9]	GC_DRNet wheat seed recognition	Self-constructed (29 varieties)	Dense residual + GCNet	96.98%	Fewer params, high accuracy
[12]	VGG16 & ResNet wheat seed ID	210 samples, 3 varieties	VGG16, ResNet50, custom CNN	98.66%	VGG16 best on small dataset

Ceyhan et al. [1] demonstrated that CNN models trained on near-infrared reflectance images of 24 Turkish wheat varieties can achieve over 99% classification accuracy, though their dataset is not publicly available. Agarwal et al. [2] showed that classical machine learning classifiers, particularly SVM, can achieve approximately 93% accuracy on handcrafted color features for binary wheat grain quality assessment. Zhao et al. [3] proposed a hybrid 1D+2D CNN for hyperspectral wheat seed classification, achieving 95.65% on eight varieties by jointly exploiting spectral and spatial information.

Mengstu and Taner [4] evaluated transfer learning on 124 wheat varieties using DenseNet201, MobileNet, and InceptionV3, with DenseNet201 achieving 95.40%. Lingwal et al. [5] trained a CNN from scratch on 15,000 images of 15 wheat grain varieties, reaching 97.53% — demonstrating that large datasets can compensate for the lack of pre-training. Xing et al. [9] proposed the GC_DRNet architecture combining dense residual connections with global context modules, achieving 96.98% on 29 wheat seed varieties. Li et al. [12] reported 98.66% accuracy using VGG16 with transfer learning on a 3-class, 210-sample wheat dataset — most directly comparable to our work.

The related work highlights that deep learning models consistently outperform traditional ML approaches for wheat seed classification. However, most studies use datasets significantly larger than ours (375 images), making direct performance comparison difficult. Our work specifically addresses the small-dataset challenge through augmentation, regularization, and attention mechanisms.

III. DATASET DESCRIPTION

We use the High-Resolution RGB Wheat Seed Dataset [10] published on Mendeley Data. The dataset contains images of three Pakistani wheat seed varieties photographed under controlled laboratory conditions using a Canon 5D Mark III camera at 256-level RGB color depth. Table II summarizes the dataset properties.

TABLE II. Wheat Seeds Dataset Properties

Property	Akbar-19	Dilkash-20	Urooj-22	Total
Images per class	125	125	125	375
Resolution	High-res RGB	High-res RGB	High-res RGB	Uniform

Background	Black (controlled)	Black (controlled)	Black (controlled)	Consistent
Image type	PNG/JPG	PNG/JPG	PNG/JPG	—
Capture device	Canon 5D Mark III	Canon 5D Mark III	Canon 5D Mark III	Same

The dataset is perfectly balanced with 125 images per class, eliminating class imbalance as a confounding factor. All images were captured against a black background under uniform lighting, ensuring consistent image quality across classes. The three varieties — Akbar-19, Dilkash-20, and Urooj-22 — represent commonly cultivated Pakistani wheat strains with subtle visual differences in shape, texture, and color, making classification non-trivial.

The dataset was split into training (70%), validation (10%), and test (20%) subsets using stratified sampling to maintain class balance across all splits. The validation set was used exclusively for monitoring training and implementing early stopping, while the test set was completely held out for final evaluation.

IV. METHODOLOGY

A. Exploratory Data Analysis

Prior to model development, exploratory data analysis (EDA) was conducted to understand the dataset characteristics. Class distribution analysis confirmed a perfectly balanced dataset, with 125 images per class. Sample images were visualized to examine intra-class variation and overall image quality. Image resolution and aspect ratio distributions were analyzed using width and height histograms with kernel density estimation. All images were verified to be in RGB format.

Additionally, an augmentation analysis was performed to determine suitable transformations for wheat seed imagery. The results indicated that horizontal flipping and mild color jitter preserve essential seed characteristics, whereas aggressive transformations such as large rotations or cropping may adversely affect shape-based features.

B. Data Preprocessing and Augmentation

All pixel values were normalized to the range [0, 1] by scaling with a factor of 1/255. For Task 2 (pre-trained baselines), images were resized to 224×224 pixels to match the input requirements of ImageNet-pretrained models. For Tasks 3 and 4 (custom models), images were resized to 64×64 pixels as a deliberate design choice to reduce model complexity and improve convergence on the small dataset.

Six augmentation strategies were applied exclusively to the training set to enhance model generalization. Table III presents the augmentation configuration and corresponding rationale.

TABLE III. Data Augmentation Configuration

Augmentation	Parameter	Rationale
Horizontal Flip	p = 1.0	Seeds appear symmetric — safe transformation
Rotation	±20°	Seeds can be oriented at any angle
Zoom	±15%	Simulates varying camera distances
Width/Height Shift	±10%	Positional variation on imaging surface
Brightness Range	[0.8, 1.2]	Accounts for lighting condition differences
Rescaling	1/255	Normalize pixel values to [0, 1]

C. Model Architectures

Five model configurations were implemented and evaluated. Table IV provides a comparative overview of all architectural decisions across tasks.

TABLE IV. Architecture Comparison Across Tasks

Component	Task 2 (Pre-trained)	Task 3 (Custom CNN)	Task 4 (Attention CNN)	Notes
Base	ImageNet weights	From scratch	From scratch + SE Block	—
Input Size	224×224	64×64	64×64	Reduced for small dataset
Conv Blocks	Frozen backbone	3× Conv/BN/ReLU/Pool	3× Conv/BN/ReLU/Pool + SE	—
Attention	None	None	SE Block (ratio=8)	After Block 2
Classifier	GAP→Dense(256)→Soft max	Dense(256)→Soft max	Dense(256)→Soft max	—
Dropout	0.5	0.5	0.5	Consistent
Optimizer	Adam (1e-4)	Adam (1e-4)	Adam (1e-4)	Consistent
Batch Size	32	16	16	Reduced for Tasks 3-4
Early Stopping	No	Yes (p=10)	Yes (p=10)	—
Augmentation	No	Yes (6 types)	Yes (6 types)	—

Task 2 — Pre-trained Baselines: Three architectures were selected based on their presence in the related work: EfficientNetB0 [11], ResNet50 [9,12], and VGG16 [12]. Transfer learning was applied by loading ImageNet weights, freezing all base convolutional layers, and adding a custom classification head consisting of GlobalAveragePooling, Dense(256, ReLU), Dropout(0.5), and Dense(3, softmax). All three models were compiled with Adam (lr=0.0001) and categorical cross-entropy loss.

Task 3 — Custom CNN: A three-block CNN was designed from scratch. Each block follows the pattern: Conv2D → BatchNormalization → ReLU → MaxPooling2D. Filter counts increase progressively (32, 64, 128) to capture increasingly abstract features. The classifier head consists of Dense(256) → BatchNormalization → ReLU → Dropout(0.5) → Dense(3, softmax). EarlyStopping with patience=10 and restore_best_weights=True was employed to prevent overfitting.

Task 4 — Attention-Enhanced CNN: The custom CNN backbone was augmented with a Squeeze-and-Excitation (SE) block [13] inserted after the second convolutional block. The SE block performs channel-wise feature recalibration through three operations: (1) global average pooling to squeeze spatial information, (2) two fully-connected layers with ratio=8 to learn channel interdependencies, and (3) sigmoid gating to scale feature maps. This mechanism enables the model to selectively emphasize informative channels while suppressing less useful ones, improving discriminative feature extraction for fine-grained wheat seed classification.

D. Training Configuration

All models were trained under identical conditions where applicable: Adam optimizer, learning rate 0.0001, categorical cross-entropy loss, 70/10/20 train/val/test split, and stratified sampling. Tasks 3 and 4 additionally employed data augmentation, batch size 16, and EarlyStopping callbacks not present in Task 2. Training and testing times were recorded for all models.

E. Evaluation Metrics

All models were evaluated using the following metrics computed on the held-out test set: Overall Accuracy, Per-class Accuracy (from confusion matrix diagonal), Weighted Precision, Weighted Recall, Weighted F1-score, Area Under the ROC Curve (AUC), Confusion Matrix, and ROC curves per class. Statistical significance of performance differences was assessed using paired t-tests on per-sample correctness arrays, with significance threshold $p < 0.05$.

V. RESULTS AND DISCUSSION

A. Task 2: Pre-trained Baselines

Table V presents the performance of the three pre-trained CNN models on the test set.

TABLE V. Pre-trained CNN Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-score	AUC	Train Time (s)	Test Time (s)
EfficientNetB0	0.3333	0.1100	0.3300	0.1700	0.5831	727.46	14.55
ResNet50	0.4762	0.5500	0.4800	0.4800	0.4966	1036.80	9.90
VGG16*	0.5565	0.6759	0.5565	0.5510	0.7990	783.04	3.86

** Best performing pre-trained model : VGG16*

VGG16 achieved the highest overall performance with 55.65% accuracy, 67.59% precision, and AUC of 0.7990 — substantially outperforming both ResNet50 (47.62%) and EfficientNetB0 (33.33%). EfficientNetB0's accuracy of exactly 33.33% — equivalent to random guessing on a 3-class problem — indicates complete class collapse, where the model predicted a single class for all inputs. This is attributed to EfficientNet%0's compound-scaled architecture being designed for large datasets and failing to converge meaningfully on only 262 training images without augmentation.

ResNet50 demonstrated moderate learning with 47.62% accuracy. Its skip connections helped prevent gradient degradation, enabling partial learning despite the small dataset. However, its 50-layer depth remains excessive for a 375-image dataset.

VGG16's superior performance, despite being the oldest architecture, can be attributed to its simpler sequential structure. With fewer skip connections and a more straightforward feature extraction pipeline, VGG16 adapts more effectively to fine-tuning on small datasets where pre-trained feature hierarchies can be directly leveraged.

The overall accuracy range of 33–56% across all three models is below expectations for a 3-class classification task. The primary limiting factors are: (1) the extremely small training set of approximately 262 images, (2) absence of data augmentation, (3) absence of early stopping, and (4) no random seed fixing affecting reproducibility.

B. Statistical Comparison — Paired T-Test

TABLE VI. Paired T-Test Results for Pre-trained Models

Comparison	t-statistic	p-value	Significant	Interpretation
EfficientNet vs ResNet	-3.4487	0.000635	Yes	Statistically Significant
EfficientNet vs VGG	-5.1646	4.14×10^{-7}	Yes	Statistically Significant
ResNet vs VGG	-2.8932	0.004063	Yes	Statistically Significant

All three pairwise comparisons yielded p-values well below the 0.05 significance threshold, confirming that the performance differences between all three models are statistically genuine and not attributable to chance. The largest t-statistic magnitude (-5.1646) was observed for EfficientNet vs. VGG, reflecting the most substantial performance gap. The ResNet vs. VGG comparison ($p=0.004063$) also confirms significant differences despite both models achieving partial learning.

C. Task 3 and Task 4: Custom and Attention CNN

Table VII presents the comprehensive comparison across all five models evaluated in this study. Results for the Custom CNN (Task 3) and Attention CNN (Task 4) are to be filled upon final training completion.

TABLE VII. Comprehensive Model Comparison — All Tasks

Model	Accuracy	Precision	Recall	F1-score	AUC	Train (s)	Test (s)
EfficientNetB0	0.3333	0.1100	0.3300	0.1700	0.5831	727.46	14.55
ResNet50	0.4762	0.5500	0.4800	0.4800	0.4966	1036.80	9.90
VGG16	0.5565	0.6759	0.5565	0.5510	0.7990	783.04	3.86
Custom CNN	0.7767	0.8184	0.7767	0.7659	0.9019	715.23	56.08
Attention CNN*	0.7812	0.8145	0.7812	0.7767	0.9197	508.12	1.67

* *Best overall model.-* Attention CNN

The Custom CNN (Task 3) was designed with 64×64 input resolution, six augmentation strategies, EarlyStopping, and Dropout(0.5) to address the overfitting challenges observed in Task 2. The reduced image size significantly decreases the parameter count, enabling more effective learning from limited training data. Training curves revealed characteristic overfitting behavior with a gap between training and validation accuracy, attributed to the validation set containing only approximately 25 images — insufficient for stable loss estimation.

The Attention CNN (Task 4) extends the Custom CNN by integrating an SE block after the second convolutional block. The SE mechanism performs adaptive channel recalibration, allowing the model to focus on the most discriminative spectral features for wheat variety distinction. This architectural enhancement improves the model's ability to capture subtle inter-class differences while maintaining computational efficiency.

Quantitatively, the Attention CNN achieved the highest overall performance with 78.12% accuracy, F1-score of 0.7767, and AUC of 0.9197, slightly outperforming the Custom CNN (77.67% accuracy). Although the performance gain in accuracy is marginal, the improvement in AUC indicates better class separability and more reliable probability estimation.

Compared to pre-trained models, both Custom CNN and Attention CNN demonstrate a substantial performance improvement of over 20% in accuracy, highlighting the effectiveness of task-specific architecture design and data augmentation when working with small datasets. While VGG16 performed best among pre-trained models (55.65%), it remains significantly inferior to the custom-designed approaches.

In terms of computational efficiency, the Attention CNN also achieved the lowest testing time (1.67 seconds), making it not only the most accurate but also the most efficient model for deployment. This is primarily due to the lightweight architecture combined with efficient feature refinement through attention mechanisms.

Overall, the results indicate that custom architectures tailored to dataset characteristics outperform deep pre-trained models in low-data scenarios, and the integration of attention mechanisms further enhances performance by improving feature representation without significantly increasing complexity.

D. Discussion

This study provides important insights into the application of deep learning models on small agricultural image datasets. Among all evaluated models, the Attention CNN achieved the best overall performance, with an accuracy of 78.12%, F1-score of 0.7767, and the highest AUC of 0.9197. It slightly outperformed the Custom CNN (77.67% accuracy), while significantly surpassing all pre-trained models. Among the pre-trained architectures, VGG16 performed best, whereas EfficientNetB0 showed the weakest performance, with accuracy equivalent to random guessing.

The observed performance differences are primarily influenced by model complexity and dataset size. Pre-trained models from ImageNet do not automatically perform well on domain-specific small datasets. Deep architectures such as EfficientNetB0 and ResNet50 require large-scale data to generalize effectively; when trained on only 375 images,

EfficientNetB0 exhibited class collapse, while ResNet50 achieved only moderate performance. In contrast, VGG16's simpler architecture allowed more effective transfer learning, making it more suitable for small datasets.

The Custom CNN addressed the limitations of small dataset size and poor generalization observed in earlier models through key design choices, including reduced input resolution (224×224 to 64×64), data augmentation, dropout regularization, and early stopping. Reducing the image size was a crucial step that decreased model complexity and enabled more effective learning from limited data. Training curves indicated mild overfitting, as training accuracy exceeded validation accuracy; however, this was effectively controlled using regularization techniques, resulting in good generalization performance.

The Attention CNN extended this architecture by integrating a squeeze-and-excitation (SE) block to enhance feature representation. This mechanism performs adaptive channel-wise recalibration, allowing the model to focus on the most discriminative features. While it did not significantly improve validation accuracy compared to the Custom CNN (75.00% vs 75.89%), it achieved a higher AUC (0.9197 vs 0.9019) and substantially reduced training time (385s vs 715s). This indicates that the attention mechanism improved feature discrimination and computational efficiency rather than raw accuracy.

These findings are consistent with prior research, where attention mechanisms such as SE blocks have shown effectiveness in improving performance on small datasets without requiring additional data. Statistical analysis using paired t-tests confirmed that differences between pre-trained models are statistically significant ($p < 0.05$), indicating that performance variations are due to architectural suitability rather than random chance.

Despite these improvements, several limitations remain. The dataset size is relatively small, restricting the learning capacity of deep models and increasing the risk of overfitting. Additionally, the small validation set may introduce variability in performance estimation.

VI. CONCLUSION

This paper presented a systematic comparative study of five CNN architectures for wheat seed variety classification on a 375-image RGB dataset. Among the pre-trained models, VGG16 achieved the best performance (55.65% accuracy, AUC 0.7990), with statistically significant superiority over ResNet50 and EfficientNetB0 confirmed through paired t-tests. EfficientNetB0 failed to learn meaningful patterns, exhibiting class collapse due to the small dataset size.

The Custom CNN (Task 3) addressed key limitations through reduced input resolution (64×64), data augmentation, dropout regularization, and early stopping, resulting in a substantial performance improvement (77.67% accuracy, AUC 0.9019). The Attention CNN (Task 4) further enhanced this architecture by integrating a Squeeze-and-Excitation block for adaptive feature recalibration. While it did not significantly improve accuracy (78.12%), it achieved the highest AUC (0.9197) and reduced training time by nearly 50%, demonstrating improved feature discrimination and computational efficiency.

The primary challenge throughout this study was the limited dataset size of 375 images, which constrained model generalization and stability. Despite this, the proposed custom architectures significantly outperformed pre-trained baselines by being better aligned with small-data conditions.

Future work should focus on: (1) expanding the dataset through additional image collection, (2) applying more advanced attention mechanisms such as CBAM or self-attention, (3) evaluating cross-dataset generalization on wheat datasets from different regions, and (4) exploring semi-supervised or few-shot learning techniques to better utilize limited labeled data.

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